STT 861: Theory of Probability & Statistics I

Chapter 4.5–4.6: Covariance, Correlation and Multivariate Distributions

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November 17, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 4, Sections 4.5 and 4.6

Roadmap: how are random variables related?

§4.2 gave an all-or-nothing answer: independent or not. Today we **measure** the strength of a linear relationship, then extend everything to n variables.

- **§4.5** Covariance, correlation, $\mathrm{Var}(aX + bY)$, the bivariate normal.
- **§4.6** Joint distributions of n variables, mutual independence, the **multinomial**.

Roadmap of Chapter 4

- 4.1–4.2 Joint, conditional, independence
- 4.3–4.4 Transformations; hierarchical models
- 4.5 Covariance & correlation ← today
- 4.6 Multivariate distributions ← today
- 4.7 Inequalities

Motivation: why a single number?

- Heights of fathers and sons, stock returns, blood pressure and age: they move **together** but not exactly.
- $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y$ only if X, Y are unrelated. A portfolio's risk depends on how its assets co-move.
- We want a number that says *which direction* and *how strongly*.

Think

If X and Y are independent, what should the measure of association be? If $Y = 2X + 1$, what should it be?

Definitions

Covariance and correlation

$$\operatorname{Cov}(X,Y) = \mathbb{E}\big[(X-\mu_X)(Y-\mu_Y)\big] = \mathbb{E}(XY) - \mu_X\mu_Y, \qquad \rho_{XY} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y}.$$

- $\operatorname{Cov} > 0$: above-average X tends to go with above-average Y. $\operatorname{Cov} < 0$: opposite.
- $\operatorname{Cov}(X,X) = \operatorname{Var} X$. Units of Cov are (units of X)$\times$(units of Y); ρ is **unit-free**.

Theorem (properties)

1. $\operatorname{Cov}(aX+b,\, cY+d) = ac\,\operatorname{Cov}(X,Y)$, hence $\rho_{aX+b,cY+d} = \operatorname{sign}(ac)\,\rho_{XY}$.
2. If X, Y are independent then $\operatorname{Cov}(X,Y) = 0$ (so $\rho = 0$).
3. $\operatorname{Var}(aX+bY) = a^2 \operatorname{Var} X + b^2 \operatorname{Var} Y + 2ab\,\operatorname{Cov}(X,Y)$.

Proof of the properties

(2) Independence gives $\mathbb{E}(XY) = \mathbb{E}X\,\mathbb{E}Y$, so $\mathrm{Cov} = \mathbb{E}(XY) - \mu_X\mu_Y = 0$.

(3) Write $W = aX + bY$, $W - \mathbb{E}W = a(X - \mu_X) + b(Y - \mu_Y)$. Square and take expectations:

$$\mathrm{Var}\, W = a^2\mathbb{E}(X - \mu_X)^2 + b^2\mathbb{E}(Y - \mu_Y)^2 + 2ab\,\mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

(1) $\mathrm{Cov}(aX + b, cY + d) = \mathbb{E}[a(X - \mu_X)\,c(Y - \mu_Y)] = ac\ \mathrm{Cov}(X, Y)$. □

Remark

Property 3 extends: $\mathrm{Var}\left(\sum_i X_i\right) = \sum_i \mathrm{Var}\, X_i + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$. For **independent** (or just uncorrelated) variables the covariance terms vanish, giving the familiar $\mathrm{Var}\,\bar{X} = \sigma^2/n$.

Example 1: computing ρ from a joint pdf

$f(x, y) = x + y$ on the unit square.

From §4.2: $f_X(x) = x + \frac{1}{2}$. So

$$\mathbb{E}X = \int_0^1 x(x + \tfrac{1}{2})dx = \tfrac{7}{12},$$

$$\mathbb{E}X^2 = \int_0^1 x^2(x + \tfrac{1}{2})dx = \tfrac{1}{4} + \tfrac{1}{6} = \tfrac{5}{12},$$

$$\operatorname{Var} X = \tfrac{5}{12} - \tfrac{49}{144} = \tfrac{11}{144}.$$

$$\mathbb{E}(XY) = \int_0^1 \int_0^1 xy(x + y)\, dy\, dx = \tfrac{1}{3},$$

$$\operatorname{Cov} = \tfrac{1}{3} - \tfrac{49}{144} = -\tfrac{1}{144}.$$

Result

By symmetry $\operatorname{Var} Y = \frac{11}{144}$, so $\rho = \dfrac{-1/144}{11/144} = -\dfrac{1}{11} \approx -0.09$: weakly **negative**, even though the density $x + y$ looks "increasing". Large x already puts weight on the density near the edge, so Y is slightly pulled down.

Example 2: $f = 2$ on $0 < x < y < 1$

Recall $f_X = 2(1-x)$ (Beta$(1,2)$), $f_Y = 2y$ (Beta$(2,1)$).

$$\mathbb{E}X = \tfrac{1}{3}, \quad \mathbb{E}Y = \tfrac{2}{3}, \quad \operatorname{Var} X = \operatorname{Var} Y = \tfrac{1}{18}, \quad \mathbb{E}(XY) = \int_0^1 \int_0^y 2xy \, dx \, dy = \tfrac{1}{4}.$$

$$\operatorname{Cov}(X,Y) = \tfrac{1}{4} - \tfrac{1}{3} \cdot \tfrac{2}{3} = \tfrac{1}{36}, \qquad \rho = \frac{1/36}{1/18} = \tfrac{1}{2}.$$

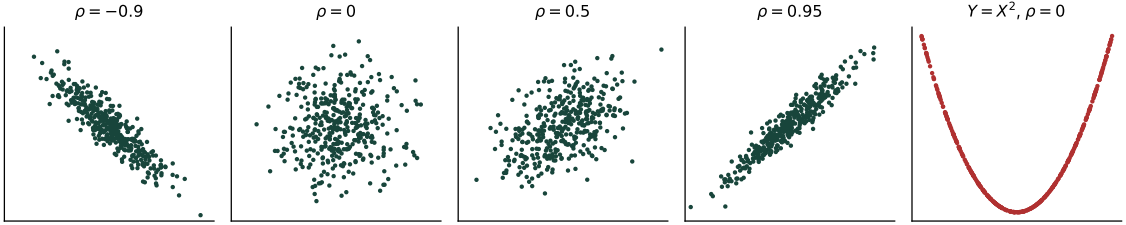
Interpretation

$X < Y$ always, so a large X forces Y to be large: positive correlation. Simulation of 10^6 pairs (max and min of two uniforms): Cov ≈ 0.02777, matching $1/36$.

Check yourself

Why is $\mathbb{E}(Y \mid X = x) = \frac{1+x}{2}$ (from §4.2) consistent with $\rho > 0$?

Correlation measures *linear* association



Independent ⇒ uncorrelated

The converse is **false**. $X \sim \mathrm{Unif}(-1, 1)$, $Y = X^2$: Y is a function of X, yet $\mathrm{Cov}(X, Y) = \mathbb{E}X^3 - \mathbb{E}X\,\mathbb{E}X^2 = 0$.

Right panel: perfectly dependent, $\rho = 0$ (simulated $\rho \approx -0.002$). Zero correlation rules out a *linear* trend only.

$-1 \le \rho \le 1$, with equality only for exact linear relations

Theorem

(1) $-1 \le \rho_{XY} \le 1$. (2) $|\rho_{XY}| = 1$ iff $P(Y = aX + b) = 1$ for some $a \neq 0$, b; $\rho = 1$ if $a > 0$ and $\rho = -1$ if $a < 0$.

Proof. Standardize: let $U = \frac{X-\mu_X}{\sigma_X}$, $V = \frac{Y-\mu_Y}{\sigma_Y}$, so $\mathbb{E}U = \mathbb{E}V = 0$, $\operatorname{Var} U = \operatorname{Var} V = 1$, $\operatorname{Cov}(U, V) = \rho$. Then

$$0 \le \operatorname{Var}(U \pm V) = 1 + 1 \pm 2\rho = 2(1 \pm \rho) \implies -1 \le \rho \le 1.$$

If $\rho = 1$, $\operatorname{Var}(U - V) = 0$, so $U = V$ with probability 1, i.e. $Y = \mu_Y + \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$: exactly linear with positive slope. $\rho = -1$ is the same with $U + V$. The converse is direct substitution. □

Equivalent form

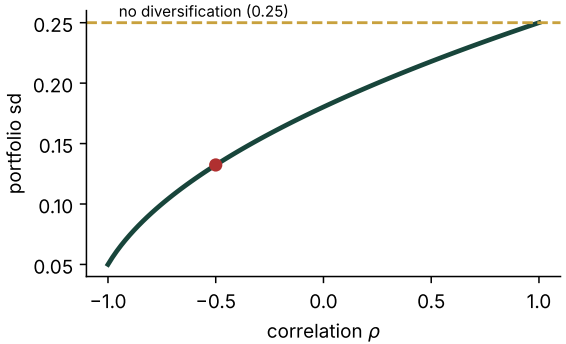
$|\operatorname{Cov}(X, Y)| \le \sigma_X \sigma_Y$ is the **Cauchy–Schwarz inequality** for random variables.

Example: portfolio risk and diversification

Two assets with $\sigma_1 = 0.2$, $\sigma_2 = 0.3$; invest 50% in each. Return $R = \frac{1}{2}R_1 + \frac{1}{2}R_2$.

$$\text{Var}\, R = \tfrac{1}{4}\sigma_1^2 + \tfrac{1}{4}\sigma_2^2 + 2 \cdot \tfrac{1}{4}\,\rho\,\sigma_1\sigma_2 = 0.0325 + 0.03\,\rho.$$

ρ	Var R	sd
1	0.0625	0.250
0.5	0.0475	0.218
0	0.0325	0.180
-0.5	0.0175	0.132
-1	0.0025	0.050



Unless $\rho = 1$, the portfolio is less risky than the weighted average of the two sds (0.25). This is **diversification**.

The bivariate normal distribution

Bivariate normal

$(X, Y) \sim N_2(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$ with $-1 < \rho < 1$ has pdf

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{ -\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}.$$

Facts

1. Marginals: $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, and $\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.
2. Conditionals: $Y \mid X = x \sim N\Big(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1-\rho^2)\Big)$.
3. $aX + bY$ is normal. **For the bivariate normal,** $\rho = 0 \iff X, Y$ **independent.**

Bivariate normal: why $\rho = 0 \Leftrightarrow$ independent, and regression to the mean

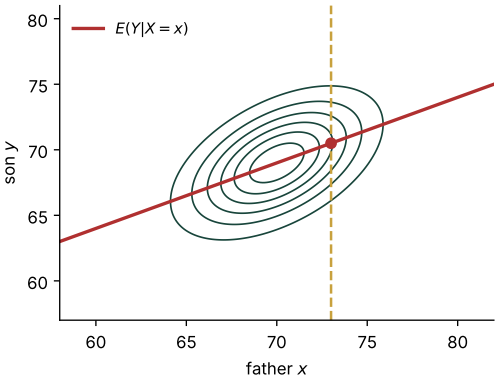
If $\rho = 0$ the exponent splits and

$$f(x,y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}}_{f_X} \underbrace{\frac{1}{\sqrt{2\pi}\sigma_Y} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}}_{f_Y}.$$

Example (fathers and sons). Heights in inches: $X \sim N(70, 3^2)$, $Y \sim N(69, 3^2)$, $\rho = 0.5$. A father is 73 in.

$$\mathbb{E}(Y \mid X = 73) = 69 + 0.5(73 - 70) = 70.5,$$
$$\mathrm{Var}(Y \mid X = 73) = 9(1 - 0.25) = 6.75.$$

$$P(Y > 72 \mid X = 73) = P\big(Z > \tfrac{1.5}{\sqrt{6.75}}\big) \approx 0.282.$$



The conditional mean moves only $\rho = 0.5$ of the way: sons of tall fathers are tall, but **closer to average**.

In-class Activity 1

Activity 1

(X, Y) has joint pmf

	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.2	0.1	0.1
$x = 1$	0.1	0.2	0.3

a. Find the marginals, $\mathbb{E}X$, $\mathbb{E}Y$, $\mathrm{Var}\, X$, $\mathrm{Var}\, Y$.

b. Compute $\mathrm{Cov}(X, Y)$ and ρ_{XY}.

c. Compute $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(X - Y)$.

d. Are X and Y independent? Justify with one cell of the table.

Activity 1: solutions

(a) $P(X=0)=0.4$, $P(X=1)=0.6$; $P(Y=0,1,2)=0.3, 0.3, 0.4$. $\mathbb{E}X=0.6$, $\mathbb{E}Y=0.3+0.8=1.1$, $\mathrm{Var}\,X=0.24$, $\mathbb{E}Y^2=0.3+1.6=1.9$, $\mathrm{Var}\,Y=1.9-1.21=0.69$.

(b) $\mathbb{E}(XY)=1(1)(0.2)+1(2)(0.3)=0.8$, so $\mathrm{Cov}=0.8-0.6(1.1)=0.14$ and $\rho=\dfrac{0.14}{\sqrt{0.24\cdot 0.69}}\approx 0.344$.

(c) $\mathrm{Var}(X+Y)=0.24+0.69+2(0.14)=1.21$; $\mathrm{Var}(X-Y)=0.24+0.69-0.28=0.65$.

(d) Not independent: $P(X=0,Y=0)=0.2\neq 0.4\cdot 0.3=0.12$ (also $\mathrm{Cov}\neq 0$).

Check

$\rho>0$ matches the table: the mass in the lower-left and upper-right corners (0.2 and 0.3) is larger than independence would give.

Motivation: from two variables to n

- A random sample is n measurements $X_1, \ldots, X_n$; statistics such as $\bar{X}$ are functions of the whole vector.
- A survey asks each of n people to pick one of k options: the counts $(X_1, \ldots, X_k)$ are **dependent** (they add to n).
- Everything in §4.1–4.5 carries over; the new ingredients are bookkeeping and one very important distribution.

Think

In a poll of $n = 100$ voters with three candidates, if candidate A gets many votes, what happens to the count for B? Is that count independent of A's?

Joint, marginal, conditional, independence in $\mathbb{R}^n$

Random vector

$\mathbf{X} = (X_1, \ldots, X_n)$ has joint pmf/pdf $f(x_1, \ldots, x_n)$. For $A \subset \mathbb{R}^n$: $P(\mathbf{X} \in A) = \int \cdots \int_A f \, dx$ (sum in the discrete case). $\mathbb{E}g(\mathbf{X}) = \int \cdots \int g \, f \, dx$.

- **Marginal** of $(X_1, \ldots, X_k)$: integrate (sum) out the other coordinates.
- **Conditional**: $f(x_{k+1}, \ldots, x_n \mid x_1, \ldots, x_k) = \dfrac{f(x_1, \ldots, x_n)}{f_{1:k}(x_1, \ldots, x_k)}$.

Mutual independence

$X_1, \ldots, X_n$ are **mutually independent** if $f(x_1, \ldots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$ for all $\mathbf{x}$. Then $\mathbb{E} \prod g_i(X_i) = \prod \mathbb{E} g_i(X_i)$, and any functions of disjoint groups are independent.

Pairwise independence does *not* imply mutual independence.

Expectation and variance of sums

Theorem

$$\mathbb{E}\Big(\sum_{i=1}^{n} a_i X_i\Big) = \sum_i a_i \mathbb{E}X_i \quad \text{(always)},$$

$$\operatorname{Var}\Big(\sum_{i=1}^{n} a_i X_i\Big) = \sum_i a_i^2 \operatorname{Var} X_i + 2\sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j).$$

Example: variance of a Binomial via indicators

$X = \sum_{i=1}^{n} I_i$ with I_i iid Bernoulli(p): $\mathbb{E}X = np$, and independence kills all covariances, so $\operatorname{Var} X = n\,p(1-p)$.

Example: Hypergeometric (dependent indicators)

Draw n of N items, M "good", without replacement; $I_i = 1$ if draw i is good. $\operatorname{Var} I_i = \frac{M}{N}(1 - \frac{M}{N})$, $\operatorname{Cov}(I_i, I_j) = -\frac{M(N-M)}{N^2(N-1)}$, so $\operatorname{Var} X = n\frac{M}{N}\big(1 - \frac{M}{N}\big)\big(1 - \frac{n-1}{N-1}\big)$: Binomial variance times the **finite-population correction**.

The multinomial distribution

Multinomial$(n; p_1, \ldots, p_k)$

n independent trials, each ending in one of k categories with probabilities $p_1, \ldots, p_k$ ($\sum p_i = 1$). X_i = number of trials landing in category i:

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!}\, p_1^{x_1} \cdots p_k^{x_k}, \quad x_i \geq 0, \ \sum x_i = n.$$

- The coefficient counts the arrangements of n trials into groups; the pmf sums to 1 by the **multinomial theorem**: $(p_1 + \cdots + p_k)^n = 1$.
- $k = 2$ is the Binomial.

Example: genotype counts

A cross yields genotypes with probabilities $(0.5, 0.3, 0.2)$. Among $n = 10$ offspring,

$$P(5, 3, 2) = \frac{10!}{5!\,3!\,2!}(0.5)^5(0.3)^3(0.2)^2 = 2520 \cdot 0.5^5 \cdot 0.3^3 \cdot 0.2^2 = 0.08505.$$

Multinomial: marginals, moments, conditionals

Theorem

1. **Marginal**: $X_i \sim \mathrm{Bin}(n, p_i)$ (category i or not). So $\mathbb{E}X_i = np_i$, $\mathrm{Var}\, X_i = np_i(1-p_i)$.
2. **Covariance**: $\mathrm{Cov}(X_i, X_j) = -np_ip_j$ for $i \neq j$, so $\rho_{ij} = -\sqrt{\dfrac{p_ip_j}{(1-p_i)(1-p_j)}}$.
3. **Conditional**: $X_i \mid X_j = x_j \sim \mathrm{Bin}\big(n - x_j,\ \frac{p_i}{1-p_j}\big)$.

Proof of 2. $X_i + X_j \sim \mathrm{Bin}(n, p_i + p_j)$, so $\mathrm{Var}(X_i + X_j) = n(p_i + p_j)(1 - p_i - p_j)$. Also $\mathrm{Var}(X_i + X_j) = np_i(1-p_i) + np_j(1-p_j) + 2\,\mathrm{Cov}$. Equate: $2\,\mathrm{Cov} = -2np_ip_j$. □

Check with $n = 10$, $p = (0.5, 0.3, 0.2)$

$\mathrm{Cov}(X_1, X_2) = -1.5$, $\rho_{12} = -\sqrt{0.15/0.35} = -0.654$. Simulation: -1.4997 and -0.654. Given $X_2 = 3$: $X_1 \sim \mathrm{Bin}(7, \frac{5}{7})$, mean 5 (simulated 5.00).

Why the covariance is negative

- The counts must add to n: $\sum_i X_i = n$ is a constant, so $\text{Var}\left(\sum X_i\right) = 0$, forcing the covariances to be negative:

$$0 = \sum_i np_i(1-p_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) = n\left[1 - \sum p_i^2\right] - n\sum_{i \neq j} p_i p_j \ \checkmark$$

- More in one category leaves fewer trials for the others: **competition for a fixed total**.
- Consequence: category counts are **never** independent, even though the n trials are.

Connection to Chapter 4.4

If $N \sim \text{Poisson}(\lambda)$ trials are made instead of a fixed n, the category counts become **independent** Poissons with means λp_i ("Poisson thinning" and the bonus question on the exam). Fixing the total creates the dependence.

Transformations in n dimensions (preview)

Change of variables in $\mathbb{R}^n$

If $\mathbf{U} = g(\mathbf{X})$ is one-to-one with inverse $\mathbf{x} = h(\mathbf{u})$, then

$$f_{\mathbf{U}}(\mathbf{u}) = f_{\mathbf{X}}\big(h(\mathbf{u})\big)\,|J|, \qquad J = \det\left(\frac{\partial x_i}{\partial u_j}\right)_{n\times n}.$$

- Same recipe as §4.3: invert, find the support, take the $n \times n$ Jacobian, then marginalize.
- Typical use: the sum $X_1 + \cdots + X_n$ of independent variables, or order statistics (Chapter 5).

Example

X_1, X_2, X_3 iid $\mathrm{Exp}(1)$, $U_1 = X_1$, $U_2 = X_1 + X_2$, $U_3 = X_1 + X_2 + X_3$ (arrival times of a Poisson process). $|J| = 1$ and $f_{\mathbf{U}} = e^{-u_3}$ on $0 < u_1 < u_2 < u_3$, so $U_3 \sim \mathrm{Gamma}(3, 1)$.

Summary

Idea	**Formula**
Covariance	$\mathrm{Cov}(X, Y) = \mathbb{E}XY - \mathbb{E}X\,\mathbb{E}Y$
Correlation	$\rho = \mathrm{Cov}\,/(\sigma_X \sigma_Y) \in [-1, 1]$; $\pm 1 \iff$ exactly linear
Variance of a combination	$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}\,X + b^2\,\mathrm{Var}\,Y + 2ab\,\mathrm{Cov}$
Independent $\Rightarrow$ uncorrelated	converse false (true for bivariate normal)
Bivariate normal conditional	$Y \mid x \sim N\big(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2)\big)$
Multinomial	$\frac{n!}{\prod x_i!}\prod p_i^{x_i}$; $X_i \sim \mathrm{Bin}(n, p_i)$; $\mathrm{Cov}(X_i, X_j) = -np_ip_j$

In-class Activity 2

Activity 2

(a) (X, Y) is bivariate normal with $\mu = (0, 0)$, $\sigma_X^2 = \sigma_Y^2 = 1$, $\rho = 0.6$. Find the distributions of $X + Y$ and $X - Y$, and compute $P(X + Y > 2)$.

(b) For the same pair, find the distribution of $Y \mid X = 1$.

(c) $n = 20$ offspring, genotype probabilities $(0.5, 0.3, 0.2)$. Write $P(X_1 = 10, X_2 = 6, X_3 = 4)$, and find $\mathrm{Cov}(X_1, X_3)$ and ρ_{13}.

(d) Find the conditional distribution of X_1 given $X_3 = 4$, and $\mathbb{E}(X_1 \mid X_3 = 4)$.

Activity 2: solutions

(a) $X + Y \sim N(0,\ 1 + 1 + 2(0.6)) = N(0, 3.2)$ and $X - Y \sim N(0,\ 2 - 1.2) = N(0, 0.8)$. $P(X + Y > 2) = P\big(Z > \frac{2}{\sqrt{3.2}}\big) = P(Z > 1.118) \approx 0.1318$.

(b) $Y \mid X = 1 \sim N(0.6 \cdot 1,\ 1 - 0.36) = N(0.6,\ 0.64)$.

(c) $P = \dfrac{20!}{10!\,6!\,4!}(0.5)^{10}(0.3)^{6}(0.2)^{4} \approx 0.0442$. $\mathrm{Cov}(X_1, X_3) = -20(0.5)(0.2) = -2$, and $\rho_{13} = \dfrac{-2}{\sqrt{5 \cdot 3.2}} = -0.5$.

(d) $X_1 \mid X_3 = 4 \sim \mathrm{Bin}\big(16, \frac{0.5}{0.8}\big) = \mathrm{Bin}(16, 0.625)$, so $\mathbb{E}(X_1 \mid X_3 = 4) = 10$.

Simulation check

(a) $P(X + Y > 2) = 0.1313$, $\mathrm{Var}(X - Y) = 0.80$ (10^6 draws). (c) simulated probability 0.0440 (exact 0.04419).

Wrap-up and next time

Remember

- ρ measures **linear** association; independence $\Rightarrow \rho = 0$, not conversely.
- $|\rho| = 1 \iff$ exact line (Cauchy–Schwarz).
- Bivariate normal: conditionals are normal with a regression line; $\rho = 0 \iff$ independent.
- Multinomial: Binomial marginals, negative covariances.

Next: §4.7

Inequalities: Cauchy–Schwarz, Hölder, Jensen, Minkowski: tools for bounding expectations.

Suggested practice: Ch. 4 exercises on covariance, bivariate normal and multinomial (numbering varies by printing).